

# A Unified Perceptual Substrate for Embodied Predictive Coding

*Toward a Physical Medium for the Free Energy Principle*

*(Position paper)*

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**Abstract** — The brain holds an internal world generated by physical priors and uses sensors to correct it where prediction and observation diverge. Model-based robotics — Kalman filters, factor-graph SLAM, particle filters, model-predictive control — already organizes perception this way, but typically with a separate state estimator per modality and the internal model held as data. This position paper proposes a different architecture. The *unified perceptual substrate* is one medium that carries both the internal world (running from priors) and the external world (entering through sensors) as configurations of one field. Same medium. Same physics. The substrate's generative model is built from physically-grounded equations in the appropriate regime — the rendering equation for light (geometric optics), free-field propagation for sound (anechoic linear regime), Newton-Euler mechanics for rigid bodies (contact regime). Internal and external content are partitioned by a Markov blanket whose sensory states are the sensor interface. Prediction errors at the blanket drive precision-weighted updates of the internal world. Under the Laplace approximation with Gaussian likelihoods, this update is gradient flow on the accuracy term of variational free energy — maximum-likelihood prediction-error minimization against an embodied generative model. The paper's central experiment, §6, demonstrates a multi-modal predictive loop closing through one substrate over eight frames; image-modality prediction error drops 59%, audio-modality residual drops 60% in waveform terms but the source-position belief moves only from 1.27 m to 1.12 m, indicating that ITD-based localization has barely begun in this run. Encoded rigid-body priors match analytical mechanics to within 0.01% on a flat slope and 0.8% on a curved arc (Appendix A: forward-model validation). The active states of the blanket, the complexity term of free energy, the Bayesian-mechanics steady-state density that would make the substrate claim formal rather than architectural, and quantitative comparison against inverse-graphics and world-model baselines are the agenda this paper opens, not the results it delivers.

## 1. Introduction

Perception is not pure reconstruction. The brain holds an internal world generated from priors about physics and uses sensors to correct that world where prediction and observation diverge [3, 4, 5]. Most of what is experienced at any moment is generated, not sensed. Sensors are the corrective tether that keeps the generated content aligned with reality.

Robotics already knows this in part. Kalman filters, factor-graph SLAM, particle filters, and model-predictive control all treat sensors as evidence that updates an internal model rather than as raw input that builds the world. The architecture we propose differs in a specific way: rather than placing each modality on its own pipeline with a separate state estimator, the internal world and the external world are treated as configurations of one medium, partitioned by a Markov blanket, with the same generative model serving every sensory channel. This is a structural proposal about how to organize the system, not a claim that we have transcended representation; like other inference systems, the current build holds parameters  $\theta$  in memory and computes predictions  $\hat{y}(\theta)$  from them.

Friston's Free Energy Principle [1, 2, 6] gives this view its most rigorous statement. A system that persists in nonequilibrium steady state with its environment can be described as minimizing a variational free energy bound on the surprisal of its sensory states. The principle decomposes into two arcs: a perceptual arc, in which beliefs about hidden states are updated to better predict sensations, and an active arc, in which actions are selected to bring sensations into line with predictions [8]. The two together constitute active inference.

The substrate the brain runs on is not abstract. Neural dynamics are oscillatory across multiple spatial and temporal scales [9, 10], and a separate line of work — McFadden's cemi field theory [13] and related proposals — argues that conscious content is itself an electromagnetic field configuration. We do not commit here to a specific biological theory of conscious content. The commitment we make is different and weaker: that one medium can be specified as the carrier of both the internal world and the external world, and that the substrate's generative model should be built from the physics of that medium at the scales of biological perception.

That choice is concrete in this paper. The substrate's generative model uses the rendering equation in the geometric-optics regime, free-field acoustic propagation in the anechoic linear regime, and Newton-Euler mechanics in the contact regime. These are physically-grounded models — radiometric and dynamical — making regime-specific idealizations: the rendering equation discards phase, diffraction, interference, and polarization; free-field propagation assumes no boundaries or reflection; rigid-body dynamics is the contact limit of continuum mechanics. They are not learned black boxes, and they are not the full field theories either. Calling them "the priors" rather than "the model of the world" is the architectural commitment: they encode known physics in a chosen regime, and inference is gradient descent against the predictions they generate.

This paper presents the perceptual arc. The substrate carries internal (prior-generated) and external (sensed) content as configurations of one medium, partitioned by a Markov blanket whose sensory states are the sensor interface. The encoded priors generate the predictions. Sensors deliver the observations. The substrate updates its internal states by precision-weighted gradient descent on prediction error — the accuracy term of variational free energy under the Laplace approximation. Three structural items are scoped out: the active states of the blanket, the complexity term of free energy, and the Bayesian-mechanics steady-state density that would convert the substrate from an architectural proposal into a formally specified physical object. All three are taken up in §7.

## **2. The Unified Perceptual Substrate**

### **2.1 The substrate is a torus**

The substrate is one medium that carries both the internal world and the external world. The internal world runs from priors. The external world enters through sensors. Both are configurations of one field. The architectural proposal is that there is one such field, not a separate state estimator per modality and a separate model of the world held as data.

Biologically, the candidate medium is the electromagnetic dynamics of neural tissue. Neural activity is oscillatory across multiple spatial and temporal scales [9, 10], and a separate line of work — the cemi field theory of consciousness and related proposals [13] — argues that conscious content is itself an electromagnetic field configuration. We borrow the structural intuition from that line of work without committing to its biological claim. The substrate's generative model is built from the physically-grounded equations introduced in §1: the rendering equation, free-field acoustic propagation, and Newton-Euler mechanics, each within its appropriate regime.

Visually, the picture is a torus: a closed shape whose body holds internal content while the region around it stands in for the external world, with the sensor interface running where the two meet. Strictly, the torus has a single connected surface, so this is an intuition pump rather than a topological claim — what the picture captures is not two surfaces but two labeled regions of one continuous medium, joined through the interface. The formal partition is the Markov blanket, given below.

Mathematically, the partition is a Markov blanket [2]. The blanket separates the states of the substrate into internal states (the agent's beliefs, parameterized by  $\theta$ ), external states (hidden causes in the environment, denoted  $s$ ), and the blanket itself — sensory states ( $s_\sigma$ ) carrying information from external to internal, and active states ( $s_a$ ) carrying information from internal to external. The sensor interface in the torus picture is the sensory portion of the blanket. Internal states update only via the blanket: predictions are tested against sensory states, and corrections flow back through the same interface.

This paper concerns the perceptual arc — the dynamics of internal states under sensory input. The active states, which would let the agent intervene on the environment, are scoped to future work (§7). Same medium. Same physics. Internal and external are configurations of one substrate, separated statistically by the blanket.

## 2.2 How the substrate operates

The substrate runs continuously. Its internal world is generated by the encoded priors — light propagation, sound propagation, motion, contact, gravity — running forward in time. The result is what the agent expects reality to look like at any moment, given everything that was known a moment ago.

Sensory states of the blanket carry information from the environment. They do not build the internal world. They enter as evidence at the region currently being attended.

The substrate predicts what sensory states should report. Reality delivers what they actually report. The discrepancy is a prediction error. Prediction error updates the internal states. Only the region currently in attention gets a significant update; the rest keeps running on priors. Over a few cycles, the attended region of the internal world aligns with the corresponding sensory evidence.

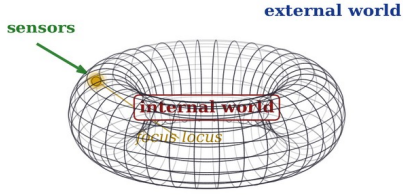
This regime — prior-driven internal dynamics with localized, precision-weighted sensory correction — is metastability in the specific sense of itinerant dynamics among quasi-attractors [11]. The substrate explores a structured landscape of semi-stable configurations; local perturbations shift the trajectory among them without collapsing the whole.

## 2.3 The focus locus as precision

Attention in the substrate is a *precision landscape*  $\Sigma^{-1}(x, t)$  over the field. Precision is the inverse variance with which the substrate weights sensory evidence against priors at each location. Where precision is high, sensory states drive large updates of the internal world. Where precision is low, the internal world keeps running on priors. The focus locus is the region where precision is currently elevated — not a token, not a binary mask, but a smooth field over the substrate.

This is the standard identification of attention with precision-weighting in active inference [12]. Optimizing  $\Sigma^{-1}$  over time is itself part of the inference problem: the substrate adjusts which regions of the field are most informative about the environment, and concentrates its corrective updates there. Asking what the agent attends to is asking where  $\Sigma^{-1}$  is currently elevated.

(a)



(b)

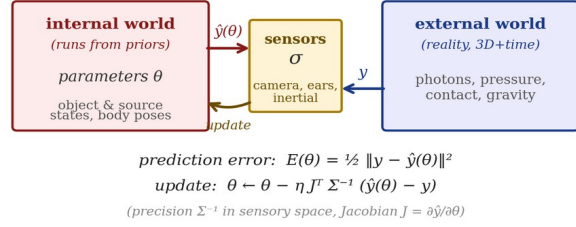


Figure 1: The unified perceptual substrate. (a) The substrate is a physical medium that carries internal and external content as configurations of one field, partitioned by a Markov blanket. The torus picture captures the intuition: internal world (label inside) and external world (label outside) are separated only by the sensor interface (sensory portion of the blanket). The focus locus (gold) marks the region of currently elevated precision. (b) Information-flow view: the internal world parameters  $\theta$  generate predicted sensor signals  $\hat{y}(\theta)$ ; the blanket delivers actual signals  $y$ ; precision-weighted prediction error  $E(\theta)$  drives the update of  $\theta$ .

### 3. Encoded Physical Priors

The priors below are physically-grounded models tied to known physics in chosen regimes — radiometric for light, free-field for sound, Newton-Euler for rigid bodies. Each subsection names the regime-specific idealization the model makes. The full Bayesian-mechanics formalization, in which the substrate's dynamics arise as gradient flow on a nonequilibrium steady-state density of the medium [2], is taken up in §7.

#### 3.1 Light

The substrate predicts what the camera sees using the rendering equation [14] — a radiometric energy-balance model in the geometric-optics regime, valid when wavelengths are much smaller than scene features and phase coherence is not needed at the sensor. For a surface point  $x$  with outgoing direction  $\omega$ , the outgoing radiance is:

$$L(x, \omega) = L_e(x, \omega) + \int_{-\Omega} f_r(x, \omega', \omega) L_i(x, \omega') (\omega' \cdot n) d\omega' \quad (1)$$

with emitted radiance  $L_e$ , reflectance function  $f_r$ , incoming radiance  $L_i$  over the hemisphere  $\Omega$ , and surface normal  $n$ . Given the internal world's belief about object geometry, materials, and light sources, integrating (1) along camera rays gives the predicted image. The observed image arrives from the camera. The pixelwise difference is the prediction error in the visual modality.

#### 3.2 Sound

The substrate predicts what the ears hear using free-field acoustic propagation — the linear-acoustic regime with anechoic boundary conditions (no reflection, no diffraction, no scattering). This is appropriate when sources are small relative to wavelength and the scene's reverberant structure is being deliberately set aside.

A source at position  $x_s$  emitting waveform  $s(t)$  produces pressure at a receiver  $x_r$  as:

$$p(x_r, t) = s(t - \|x_s - x_r\|/c) / (4\pi \|x_s - x_r\|) \quad (2)$$

where  $c$  is the speed of sound. For two ears at  $x_l$  and  $x_r$ , the interaural time difference is:

$$ITD = (\|x_s - x_l\| - \|x_s - x_r\|) / c \quad (3)$$

The substrate predicts left and right ear signals from its belief about source position. Cross-correlation of the two channels gives the observed ITD. The difference updates the substrate's belief about where the source is.

### 3.3 Rigid body motion

Matter in the internal world moves according to Newton-Euler dynamics — the contact regime of classical mechanics, valid at the speeds and length scales of perceived motion, treating bodies as rigid (no deformation) with contact resolved as constraint impulses.

For a body of mass  $m$  and inertia tensor  $I$ :

$$m \dot{v} = F, \quad I \dot{\omega} + \omega \times (I \omega) = \tau \quad (4)$$

Translation is integrated by velocity Verlet, which is exact for constant acceleration:

$$x(t+\Delta t) = x(t) + v(t) \Delta t + \frac{1}{2} a(t) \Delta t^2, \quad v(t+\Delta t) = v(t) + a(t) \Delta t \quad (5)$$

Rotation uses symplectic Euler with quaternion exponential to preserve unit norm [15]:

$$\omega(t+\Delta t) = \omega(t) + I^{-1}(\tau - \omega \times (I \omega)) \Delta t, \quad q(t+\Delta t) = \exp(\frac{1}{2} \omega \Delta t) \cdot q(t) \quad (6)$$

### 3.4 Contact and friction

When a body touches a surface, contact is resolved by impulse constraints. The non-penetration condition requires the post-impulse normal velocity to be non-negative. For a contact at body-frame offset  $r$  with surface normal  $\hat{n}$ , the effective mass along the normal is:

$$1/m_{eff} = 1/m + (r \times \hat{n})^T I^{-1} (r \times \hat{n}) \quad (7)$$

Multiple simultaneous contacts (the four wheels of a skateboard, for example) couple through the rigid body. They are resolved by iterative Gauss-Seidel relaxation. Coulomb friction bounds the tangential impulse:

$$|J_t| \leq \mu J_n \quad (8)$$

with friction coefficient  $\mu$ . Equations (1)–(8) constitute the substrate's generative model of physical reality.

## 4. The Predictive Loop and the Free Energy Principle

### 4.1 The loop

Let  $\theta$  be the parameters of the internal states: object positions, sizes, source locations, body states. The encoded priors generate predictions  $\hat{y}(\theta)$  of what the sensory states of the Markov blanket should report. The blanket delivers actual sensor observations  $y$ . Let the squared prediction error be:

$$E(\theta) = \frac{1}{2} \|y - \hat{y}(\theta)\|^2 \quad (9)$$

Internal states update by precision-weighted gradient descent on the prediction error:

$$\theta(t+1) = \theta(t) - \eta \cdot J^T \Sigma^{-1}(t) (\hat{y}(\theta(t)) - y) \quad (10)$$

with learning rate  $\eta$ , precision matrix  $\Sigma^{-1}(t)$  — the inverse of the observation noise covariance, acting in sensory space — and Jacobian  $J = \partial \hat{y} / \partial \theta$  carrying the precision-weighted residual from sensory space back into parameter space. The precision plays the role of attention: regions of sensory space with high precision drive large updates of the corresponding parameters; regions with low precision keep running on priors [12]. The

focus locus is the region of currently elevated precision. Over a few cycles, the high-precision region of the internal world aligns with the corresponding sensory evidence.

## 4.2 Correspondence with variational free energy

The Free Energy Principle [1, 2] states that a self-organizing system minimizes variational free energy:

$$F = E_{q(s)}[-\log p(y, s)] - H[q(s)] \quad (11)$$

with approximate posterior  $q(s)$  over hidden states  $s$ , joint density  $p(y, s) = p(y|s) p(s)$  given by the generative model, and entropy  $H$ .  $F$  upper-bounds the surprisal  $-\log p(y)$ , so minimizing  $F$  maximizes model evidence. Free energy decomposes as:

$$F = E_{q(s)}[-\log p(y|s)] + D_{KL}[q(s) \parallel p(s)] \quad (12)$$

The first term is the accuracy: how well the model's predictions match the observations. The second term is the complexity: how far the approximate posterior departs from the prior. Minimizing both is what protects inference from overfitting the sensorium.

Under the Laplace approximation,  $q(s)$  is taken to be Gaussian centered on a MAP estimate  $s^*$  (in our notation,  $\theta$ ), with the likelihood  $p(y|s)$  also Gaussian with precision  $\Sigma^{-1}$ . The accuracy term then reduces to the precision-weighted squared prediction error [7, 16]:

$$E_{q(s)}[-\log p(y|s)] \approx \frac{1}{2} (y - \hat{y}(\theta))^T \Sigma^{-1} (y - \hat{y}(\theta)) + \text{const} \quad (13)$$

Taking the gradient of (13) with respect to  $\theta$  yields  $J^T \Sigma^{-1} (\hat{y}(\theta) - y)$ , where  $J = \partial \hat{y} / \partial \theta$  is the Jacobian of the prediction with respect to the parameters. This is precisely the update direction in equation (10): the precision weights the residual in sensory space, and the Jacobian transpose carries the result back into parameter space. Because the substrate's update descends only on the accuracy term and does not include a prior-gradient contribution  $\nabla_{\theta} \log p(\theta)$ , the procedure is honestly described as *maximum-likelihood prediction-error minimization* — equivalent to MAP estimation only under a flat prior on the parameters. This is the perceptual arc of the Free Energy Principle in the Laplace approximation, with the complexity term held implicit by treating priors as known physics. The internal state parameters  $\theta$  play the role of the point estimate of hidden states  $s$ . The encoded physical priors play the role of the likelihood structure in the generative model  $p(y, s)$ . Restoring  $\nabla_{\theta} \log p(\theta)$  to the update would re-introduce the complexity term and produce a genuine MAP estimate; this is one of the natural extensions discussed in §7.

Two structural complements are not addressed in this paper. First, the complexity term: a complete account of free energy minimization would let  $q(s)$  depart from the priors in a regularized way, allowing the substrate to learn priors rather than treat them as fixed. Second, the active states of the Markov blanket: a complete active inference agent would also act on the environment to bring observations into line with predictions. Both are open directions, and we return to them in §7.

The architecture is a candidate realization of the perceptual arc: a generative model built from known physics in specified regimes, run as the engine that produces predictions against which sensor evidence is compared. The current build holds  $\theta$  in memory and computes  $\hat{y}(\theta)$  from it. Making the substrate's update a gradient flow on a steady-state density of the medium itself — rather than a program that computes such a flow — is the formal step we have not taken, and we are explicit about that in §7.

## 5. Related Work

The architecture proposed here sits between several literatures. We name them explicitly because the closest neighbors are the natural source of confusion.

Predictive coding and active inference. The framework of §4 is standard predictive coding [4, 5] interpreted through the Free Energy Principle [1, 2, 6, 8]. What is non-standard is the commitment to a single medium carrying both internal and external content (§2.1) and the use of known physics rather than learned generative models. The mathematical machinery — Markov blankets, Laplace approximation, precision-weighted prediction error — is shared with the broader active-inference literature [7, 12, 16].

Inverse graphics and analysis-by-synthesis. Section 6's loop-closing experiment is, in form, gradient-based inverse rendering: a parameterized scene generates a predicted image, the residual against the observed image drives gradient descent on the parameters. Inverse graphics with differentiable renderers is a mature literature with strong empirical results. The architectural distinction is in scope and commitment: inverse-graphics systems typically treat the renderer as a tool for fitting a particular image-formation model, and they are usually visual-only. The substrate proposal applies the same gradient-update mechanism to a single internal world that also generates audio and rigid-body predictions, all through one Markov blanket. Quantitative comparison against differentiable-rendering baselines on the same scene is the natural empirical follow-on; we do not run such a comparison here, and we flag this as the most important missing experiment.

Differentiable physics. The use of velocity-Verlet and a contact solver in §3.3 places this work adjacent to differentiable-physics simulators developed for learning dynamics and for trajectory optimization. Our priors are hand-coded rather than learned, and the gradient is used to update beliefs about world state rather than to learn the dynamics. Differentiable physics is a candidate baseline for the rigid-body half of the substrate's predictions, in the same way that differentiable rendering is the natural baseline for the visual half.

World models in machine learning. Recent work on world models [17, 18, 19] learns a generative model of an environment from interaction data and uses it for planning and policy learning. Our architecture inverts the data-versus-physics axis: we encode known physics as the model rather than learning the model from data, and we do not yet learn priors from experience. World models are the closest analog to the substrate's role in the larger architecture; the difference is what supplies the model. A hybrid in which physics priors are the base layer and learned components handle the residual — particularly for conceptual content not covered by known physics — is the more ambitious version of this work and is sketched in §7.

Bayesian mechanics and the free energy principle for a particular physics. The strongest version of the substrate claim — that the medium has a nonequilibrium steady-state density whose gradient flow is the substrate's dynamics — is precisely the program of Bayesian mechanics [2]. We make no formal contribution to that program here. The substrate as we present it is architectural; making it formal is the topic of §7.

## 6. Experiment: predictive loop closing

The paper's central experiment demonstrates the predictive loop of equation (10) closing across two sensory modalities. This is loop-closure in action — not a Bayesian model comparison, not a quantitative baseline study against inverse-graphics or world-model systems. Both of those are flagged in §5 as the natural empirical follow-ons. The current §6 establishes that the loop closes; subsequent work will measure how well it closes relative to alternatives. Forward-model validation of the encoded priors against analytical mechanics is reported in Appendix A.

The substrate runs an internal world with two spheres on a textured surface under a point light. A sound source in the scene emits a tone. A  $320 \times 240$  camera observes the scene. Two microphones, 17 cm apart, sample the audio. The internal world starts with parameters that are offset from reality: sphere positions, sizes, and colors are wrong; the sound source position is offset. At each frame the substrate predicts the camera image from (1), predicts the ear signals from (2)–(3), computes the squared prediction error (9) across both modalities, and updates the internal states by (10) with constant precision  $\Sigma^{-1} = \mathbf{I}$ .

Over eight frames, total prediction error drops by 65%. The image-modality prediction error drops from a mean of 19.6 to 8.1 — a 59% reduction. The audio-modality residual drops by 60% in waveform terms (RMS of the prediction error in the recorded pressure signal), but this is the easier half: matching amplitude and rough timing of the tone explains most of that drop. The harder quantity — the substrate's belief about the source position — moves only from 1.27 m error to 1.12 m, about 12%. ITD-based localization has barely begun in this run. The audio loop is in early stages of localizing the source, even though the waveform residual has dropped sharply. We report both numbers so that the 60% headline is not misread.

The substrate is multi-modal because it is one architecture that physical reality enters through different sensors, not because the substrate has done anything clever to bind across modalities. There is no separate visual encoder and audio encoder; there is no learned cross-modal binding. This is a structural property of the architecture, not an empirical result — any joint generative model over multiple observation channels has it. The empirical claim is narrower: the loop closes through one substrate, and visual and audio predictions improve together under the same update rule.

The experiment is a single run, eight frames, with learning rate and precision fixed by hand and  $\Sigma^{-1} = \mathbf{I}$ . It is not a Bayesian model comparison; it does not report variance across multiple runs; it does not compare against a particle filter, a learned encoder, or a differentiable-rendering baseline on the same scene. Each of these is a natural next step, and we name them rather than gesture at them: (i) multiple runs with random initial parameter offsets, reporting variance; (ii) a particle-filter or Kalman baseline on the same camera observations, with identical priors; (iii) a learned-encoder comparison; (iv) an experiment in which  $\Sigma^{-1}$  is spatially structured rather than identity, exercising the precision machinery introduced in §2.3 and §4.1.

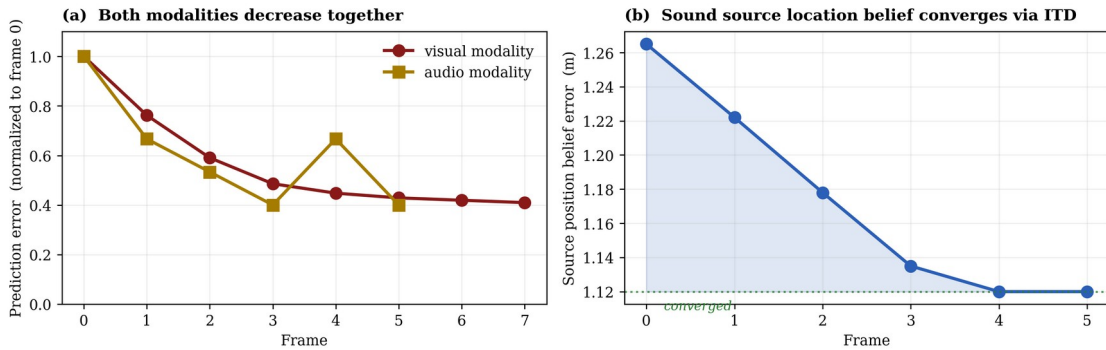


Figure 2: Multi-modal predictive loop closing through one substrate. (a) Prediction error per frame normalized to frame 0 for both modalities. Visual (red) and audio (gold) decrease together because they are driven by the same substrate-level update rule (10). (b) The substrate's belief about sound source position converges from 1.27 m error to 1.12 m via interaural time difference.

## Appendix A. Forward-model validation

The encoded priors of §3.3 (Newton-Euler dynamics, velocity-Verlet integration, symplectic-Euler rotation, iterative contact resolution) are textbook physics correctly assembled. To establish that the substrate's forward



model integrates that physics with the expected accuracy — separately from any question about the inference machinery in §6 — we run the substrate forward with no sensor corrections and compare against closed-form analytical solutions in two scenarios. This is validation of the priors as a physics engine. It is not evidence about inference, perception, or free energy.

### A.1 Skateboard on a flat slope

A skateboard (80 cm × 4 cm × 20 cm, 2 kg) starts at rest on a 20° slope with kinetic friction coefficient  $\mu = 0.2$ . The substrate predicts the descent from the rigid body priors (4)–(8). The analytical acceleration along the slope is  $a = g(\sin 20^\circ - \mu \cos 20^\circ) = 1.5115 \text{ m/s}^2$ , so the speed at time  $t$  is  $v(t) = a t$ .

| Time (s) | Simulated speed (m/s) | Analytical (m/s) | Error (%) |
|----------|-----------------------|------------------|-----------|
| 0.333    | 0.5039                | 0.5038           | 0.005     |
| 0.667    | 1.0077                | 1.0077           | 0.002     |
| 1.000    | 1.5116                | 1.5115           | 0.002     |
| 1.500    | 2.2673                | 2.2673           | 0.001     |
| 1.967    | 2.9727                | 2.9727           | 0.001     |

Table 1: Simulated speed against analytical prediction. Error remains below 0.01% across two seconds.

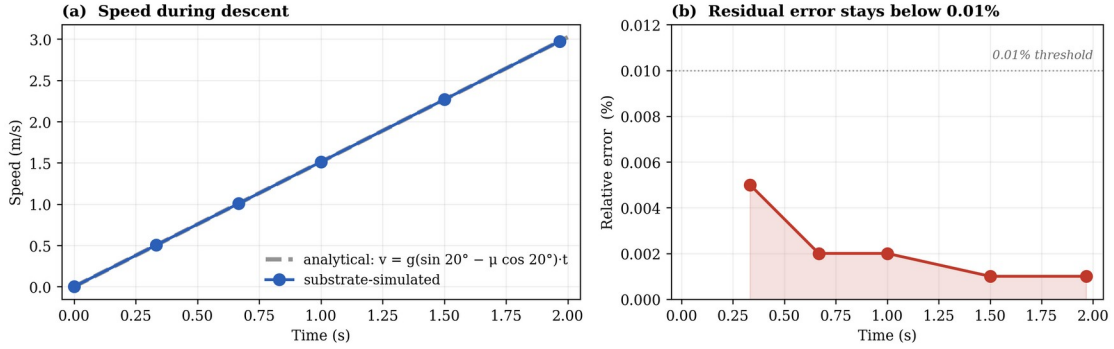


Figure 3: Skateboard on a flat 20° slope with friction  $\mu = 0.2$ . (a) Simulated speed (blue) overlaps the analytical prediction  $v(t) = g(\sin 20^\circ - \mu \cos 20^\circ)t$  (gray dashed). (b) The residual relative error stays below the 0.01% threshold across the full two-second descent, with a maximum of 0.005% at the first sample point.

### A.2 Skateboard on a curved arc

A four-wheeled skateboard starts at  $\varphi = 25^\circ$  on a circular arc slope of radius  $R = 3 \text{ m}$ . The wheels are the contact points; the deck hovers 2.5 cm above the surface. This scenario tests position-dependent surface normal, multi-point contact through the four wheels, and continuous reorientation of the board as the surface curves. The analytical speed from energy conservation in the frictionless case is:

$$v(\varphi) = \sqrt{2 g R (\cos \varphi - \cos \varphi_{\text{start}})} \quad (14)$$

| Time (s) | $\varphi$ (°) | Simulated $v$ (m/s) | Analytical $v$ (m/s) | Error (%) |
|----------|---------------|---------------------|----------------------|-----------|
| 0.133    | 24.29         | 0.5476              | 0.5519               | 0.77      |
| 0.267    | 22.19         | 1.0657              | 1.0740               | 0.77      |
| 0.400    | 18.83         | 1.5256              | 1.5375               | 0.78      |

|       |       |        |        |      |
|-------|-------|--------|--------|------|
| 0.533 | 14.39 | 1.9006 | 1.9155 | 0.78 |
| 0.667 | 9.11  | 2.1674 | 2.1847 | 0.79 |

Table 2: Simulated speed against analytical prediction on the curved arc. The error sits consistently around 0.77-0.79% — flat across the descent rather than growing, which is the signature of a small systematic bias in the discrete contact-normal handling at each timestep rather than accumulating numerical noise. The bias stays below the 0.8% threshold across the validated range.

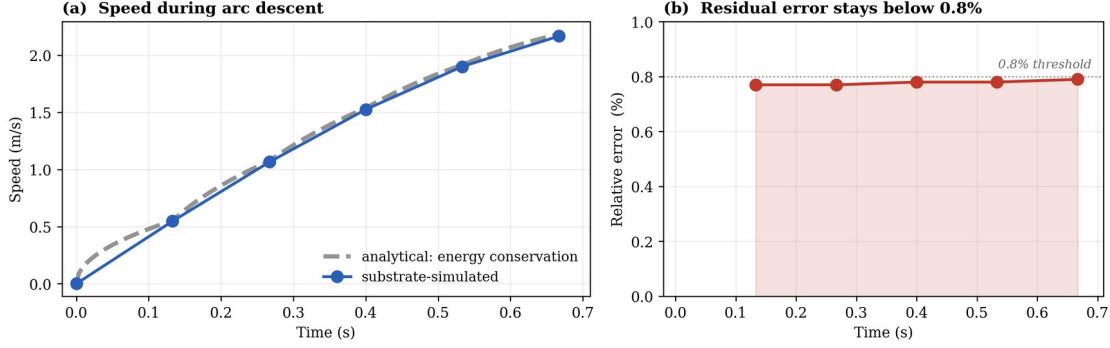


Figure 4: Skateboard on a curved arc slope, multi-point wheel contact. (a) Simulated speed (blue) follows the analytical prediction from energy conservation (gray dashed). (b) The residual relative error stays consistently below the 0.8% threshold across the validated range of the descent.

## 7. Discussion

Three results, scoped honestly. First, a unified perceptual substrate is implementable as an architectural proposal. One medium carries internal and external content as configurations of one field, partitioned by a Markov blanket whose sensory states are the sensor interface. The substrate is multi-modal by construction because all modalities enter the same medium through the same blanket — no separate encoders, no learned cross-modal binding. This is a structural property of the architecture, not an empirical advantage demonstrated against baselines. Second, the encoded priors integrate correctly as physics in their stated regimes. Forward integration matches analytical mechanics to within 0.01% on a flat slope and 0.8% on a curved arc with multi-point contact (Appendix A). This validates the priors as a physics engine in their chosen regime; it is not evidence about inference. Third, the substrate's update rule is the perceptual arc of the Free Energy Principle in the Laplace approximation. Precision-weighted gradient descent on prediction error performs maximum-likelihood estimation of the internal states under known priors. Restoring the prior-gradient term promotes the update to genuine MAP and re-introduces the complexity term in one step.

Four things remain to be done. The paper would mislead by understating any of them, and the absence of the first two means this is properly a position paper rather than an empirical contribution.

Comparative empirical study. The loop-closing experiment in §6 demonstrates that the predictive loop closes — but "the substrate works" requires comparison against the natural baselines: a particle-filter or Kalman variant on the same observations with identical priors; a learned-encoder pipeline; differentiable rendering with the same physics priors. None of these are run here. Each is straightforward to implement on the same scene, and each would convert the architectural claim into an empirical one.

Action. The Free Energy Principle has two arcs. We have implemented the perceptual one. A complete agent would also use the active states of the blanket to act on the environment, bringing sensations into line with

predictions [8]. This is half the principle. Adding active states is the natural next step: the same  $\theta$  updated by (10) would also drive action variables, and expected free energy would replace plain accuracy as the quantity minimized over policies.

Complexity. Equation (10) descends on the accuracy term. The complexity term — the divergence between the posterior and the prior — is held implicit by treating priors as fixed physics. This is appropriate for the architectural claim that physics priors are not learned from data, but it is not full variational inference. A substrate that learns its priors, or that regulates precision  $\Sigma^{-1}$  as part of the inference, would minimize free energy proper and gain the Occam character that complexity provides.

Steady-state density. The strongest version of the substrate claim is that the medium has a nonequilibrium steady-state density whose gradient flow is the substrate's own dynamics — Bayesian mechanics on a physical field [2]. The priors in §3 are physically-grounded models in chosen regimes; the formal step we have not taken is to exhibit the medium itself as a NESS density and show that the substrate's update is a gradient flow on that density. That would convert "the update resembles free energy minimization" into "the update is free energy minimization on a specified physical object." This is the form in which the substrate claim becomes load-bearing rather than architectural, and is the next paper.

Beyond the structural items, the current build is small. Conceptual content — abstract concepts, language, memory — is not yet realized in the substrate, but corresponds to persistent or hierarchical configurations of the same medium and admits incremental development, possibly through hybrid architectures in which physics priors are the base layer and learned components handle the residual.

The convergence with the Free Energy Principle is structural. Friston's framework gives the mathematical scaffold and the steady-state physics. The substrate gives a candidate architectural proposal for an embodied medium in which that scaffold can run. The work of joining the two — quantitative comparison against neighboring methods, exhibiting the steady-state density, restoring the complexity term, adding the active states — is the agenda this paper opens, not the result it delivers.

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